

# JAKUB KIERZNOWSKI

Junior Software Engineer | Software Developer Intern at IBM  
CS & Intelligent Systems Student at AGH

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## SUMMARY

Backend engineer working in Java and Python. At IBM I optimise graph and tree traversal in Data Lineage over enterprise metadata; the same algorithmic work runs through DrugPath, an agent over a 47,031-node biomedical graph, and CarCharging, a sliding window over half-hourly forecasts. After hours I build and support InstalDesk, a live B2B SaaS with a native integration to the Polish KSeF e-invoicing system. Final-year Computer Science and Intelligent Systems at AGH, GPA 4.74/5.00, graduating January 2027.

Java 21 · Spring Boot · Spring Security · PostgreSQL · Neo4j · Python · FastAPI · Docker · RabbitMQ and MQTT · Jenkins · GitHub Actions

## EXPERIENCE

### Software Developer Intern

IBM Software Lab · Kraków · hybrid  
Feb 2026 – Present

- Java and Spring on Data Lineage: optimise performance over large enterprise metadata graphs by implementing graph and tree traversal algorithms. (Client data and exact figures are under NDA.)
- Cut the pipeline test run from 20 minutes to 4 by making the automated pytest suite asynchronous — parametrised cases, session-scoped fixtures.
- Own the Jenkins job scheduling for the pipeline.
- Daily Scrum in an English-speaking cross-functional team.

### IT Team Lead, Web & Workflow Automation

Student Council · WRSS WEALiIB, AGH  
Mar 2025 – Present

Council member since Mar 2024, leading its IT team since Mar 2025. I maintain the council website (Vue, TailwindCSS) and the VPS it runs on, and built internal automations on self-hosted n8n against Slack and Google APIs over OAuth 2.0. Event coordinator for the VOLT 2025 tech conference, eight partner companies including Aptiv, PwC Polska and Cytiva, and part of the organising team for VOLT 2026.

### Team Mentor and Community Leader

Zwolnieni z Teorii · remote  
Sep 2022 – May 2025

Mentored 35+ student teams per edition, from planning to the final presentation of their social-impact projects. Later ran a participant community (ZWzT × PwC certificate) and supported olympiad entrants.

### Game Developer

AGH Code Industry · Dec 2023 – Jun 2024

Gameplay mechanics in Unity and C# for “Student Trainer”, a game built by a student team under a mentor.

## PROJECTS

Code at [<repo>](https://github.com/qualv13)

### InstalDesk, B2B SaaS for Polish installation contractors

instaldesk.pl · live since Apr 2026, v1.9.0, pilot users · closed source · Python, FastAPI, PostgreSQL, SQLite, Stripe, KSeF API

- Native KSeF integration: API v2 against the FA(3) schema, KSeF-token auth on web, XAdES-BES certificate auth on desktop — filed straight to the Ministry of Finance, without a third-party integrator.
- A web PWA and a Windows desktop client share one account. PostgreSQL in the cloud, SQLite locally so the desktop client keeps working offline.
- Per-tenant isolation, KSeF tokens encrypted at rest. Stripe billing over card, BLIK and Przelewy24, a 14-day trial without a card, renewal flows, transactional email, PDF generation, XLSX and CSV import and export.
- Built and supported solo, including user contact.

### IoT fleet backend

IoTServerApp · Java 17, Spring Boot 3.3, PostgreSQL, RabbitMQ/MQTT, Protocol Buffers, Docker

Layered Spring backend for managing smart-lamp fleets. JWT auth with refresh tokens, MQTT command and telemetry topics over RabbitMQ, a Protocol Buffers device protocol, OTA firmware delivery from S3, scheduled device jobs. Full Docker Compose stack.

## DrugPath, biomedical knowledge-graph agent

neo4j-agent · Neo4j Aura Agent Hackathon 2026, top 6 · Neo4j Aura, Cypher, Python, LLM agents

An agent that walks the Hetionet biomedical graph (47,031 nodes, 293k relationships) along drug → gene → pathway → disease paths to explain how a drug works and where it might be repurposed. Educational project on public research data. Demo at [qualv13.github.io/neo4j-agent](https://qualv13.github.io/neo4j-agent).

## BPM-first workout playlist generator

fit2beat.toadres.pl · HackYeah 2025, top 8 of 65+ · Python, Flask, Docker

Builds playlists from a target BPM rather than from genre. Nobody else in the biohacking track approached it that way. Team of four, 24 hours, 3.9/5 from the jury.

## Green-window EV charging optimiser

CarCharging · Java, Spring Boot, REST, Docker

Finds the cleanest 1–6 hour charging window inside the next 48 hours from the UK Carbon Intensity API, with a sliding window over half-hourly generation slots, and serves it and the daily fuel mix through a REST API.

## Graph and deep neural networks

GGSN · Graph and Deep Neural Networks, AGH coursework · PyTorch, TensorFlow, scikit-learn

Graph neural networks (GCN, GAT, GraphSAGE, GIN) over the Yelp dataset, benchmarked against one another. Transfer learning with Grad-CAM explanations and DistilBERT sentiment classification. Network compression by pruning, quantisation and distillation across seven architectures, compared on accuracy against model size.

## SKILLS

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### Tech

**Core:** Java 21, Spring Boot, Spring Security, Hibernate/JPA, Maven, JUnit 5, PostgreSQL, SQL, REST APIs, OOP and design patterns, algorithms and data structures (graph and tree traversal, sliding window, complexity analysis).

**Working knowledge:** Python (FastAPI, pytest), Docker and Compose, Linux/Bash, Git, Jenkins and GitHub Actions CI/CD, RabbitMQ, MQTT, Protocol Buffers, Testcontainers, OAuth 2.0, Neo4j and Cypher, AWS S3.

**Also used:** Vue with TailwindCSS, C# (Unity), C++, n8n, LLM APIs, self-hosted models (Ollama), TensorFlow, scikit-learn.

### Soft

Agile and Kanban on GitHub project boards at IBM, Jira before that; daily Scrum in a cross-functional English-speaking team. Lead event coordinator for the VOLT 2025 conference, including partnership deals with tech companies. I ship my own product and handle its support and user contact myself.

Languages: Polish native, English C1 (working language at IBM), French A2.

## EDUCATION

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### Computer Science and Intelligent Systems, BEng

AGH University of Krakow · Kraków  
Oct 2023 – Jan 2027

GPA 4.74/5.00. Year Representative since the 1st semester, Student Council since the 2nd. Engineering thesis: applying deep learning to 2D-image to 3D-mesh conversion.

### Extended Mathematics, Physics and Computer Science

V High School · Kraków  
2019 – 2023

C++, competitive-programming problem solving.

## CERTIFICATIONS

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Neo4j Certified Professional and Building Agents in Neo4j Aura, both 2026.

Amazon Junior Software Developer (Coursera specialisation): Data Structures and Algorithms, Programming with Java, Application Development.

## INTERESTS

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Snowboarding, sailing, a self-hosted homelab with Docker and quantised local models on a Raspberry Pi and a VPS, game dev, board games.